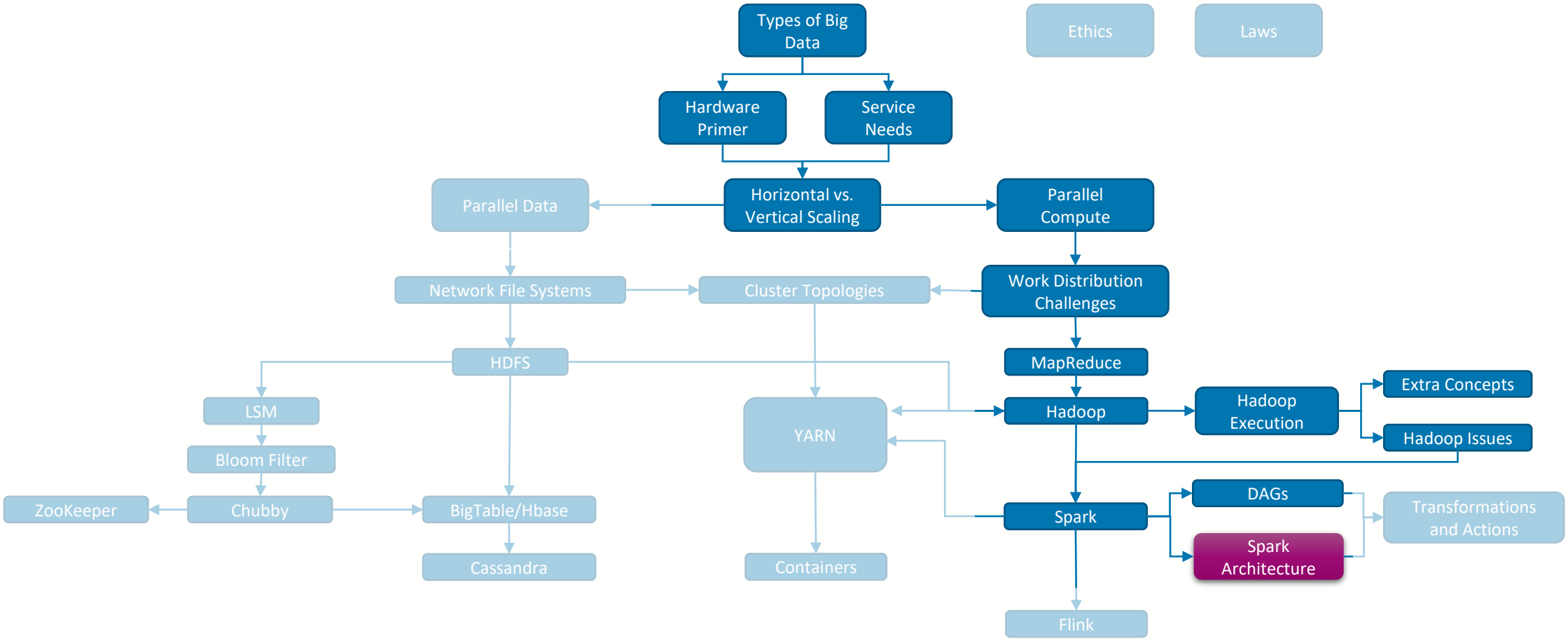
W3-06: Spark Architecture

COMPSCI4064 (H) and COMPSCI5088 (M)

Dr. Richard McCreadie

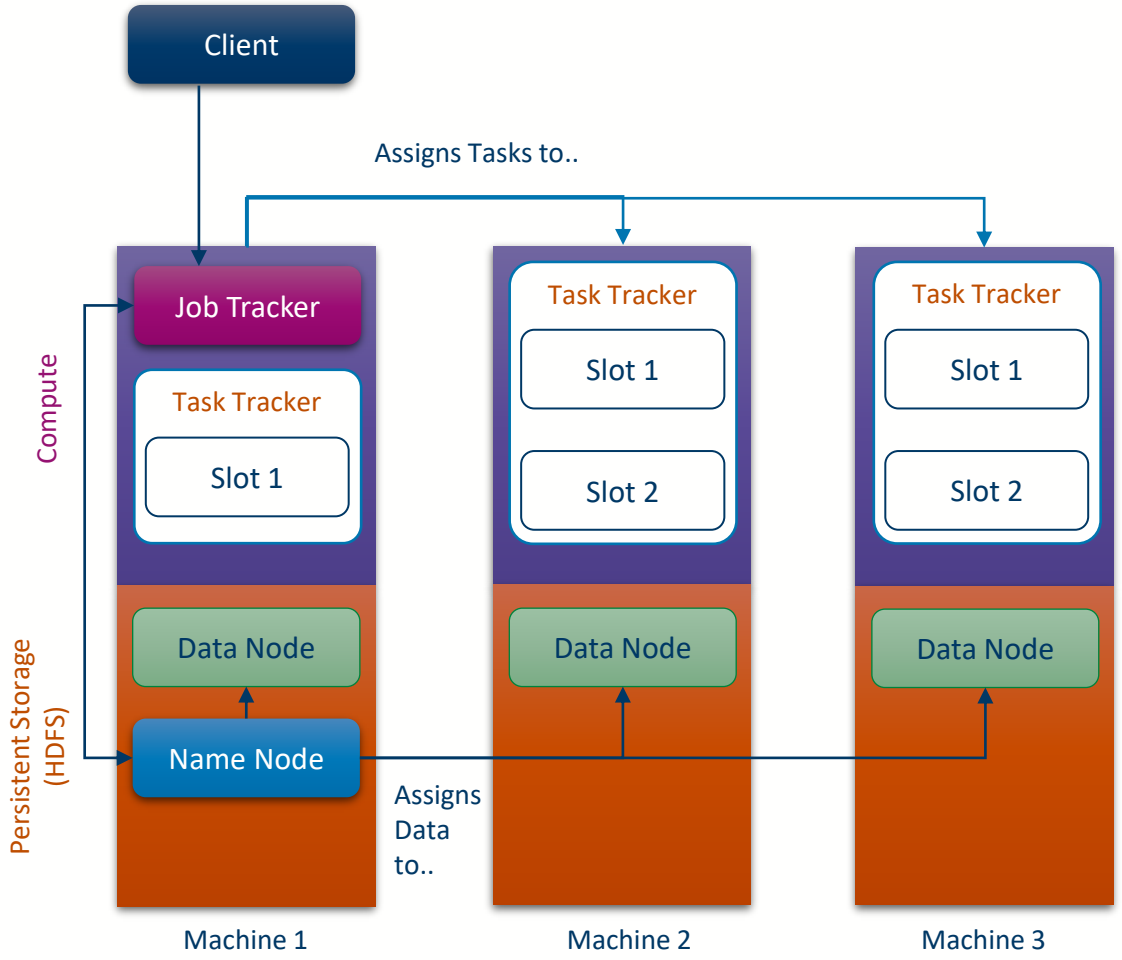
University of Glasgow

Where are we?



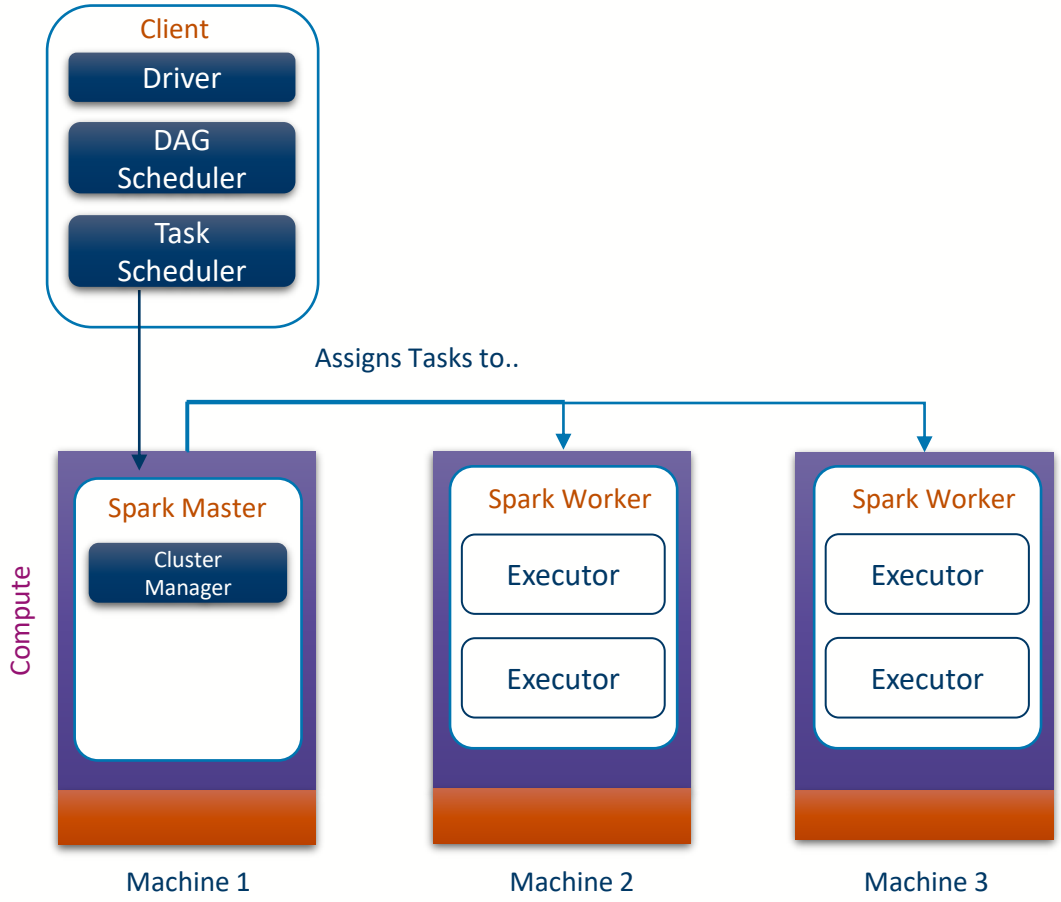
Recall the Hadoop Architecture…

- Hadoop Compute
  - Client: Software that the user uses to submit work to the Hadoop cluster, communicates with the Job Tracker
  - Job Tracker: Receives work from the client, uses the Name Node to identify where data is located, assigns work to task trackers
  - Task Tracker: Executes Map and Reduce tasks
- Hadoop HDFS
  - Name Node: Tracks the location of all data held on the HDFS, handles replication
  - Data Node: Handles storage and access of the local files on the machine



Spark Architecture

- Spark Compute
  - Client: Software that the user uses to submit work to the Spark cluster, communicates with the Spark Master
  - Driver: The program that executes the Spark Job, may run on the Client or on a node in the cluster
  - Spark Master: Receives work from the client, analyses the DAG assigns work to task trackers, communicates with the Driver
  - Spark Worker: Manages the local execution of Transformations and Actions
  - Executor: A JVM that executes tasks using a ThreadPool
    - Responsible for executing application code for a specific task/stage, caching datasets/results



University of Glasgow

Important Differences

- No defined persistent storage layer
  - Spark may use temporary storage if RAM is exhausted
  - Application code defines how data is loaded
- The Driver program must remain alive until the Spark Job finishes (since it will be the subject of the final Action)
- The Driver does not need to be run on the Client
- Each Job gets its own Executer on the Workers (to avoid Jobs sharing memory), an Executer can run multiple tasks in parallel
  - Much faster to get a job going than in Hadoop

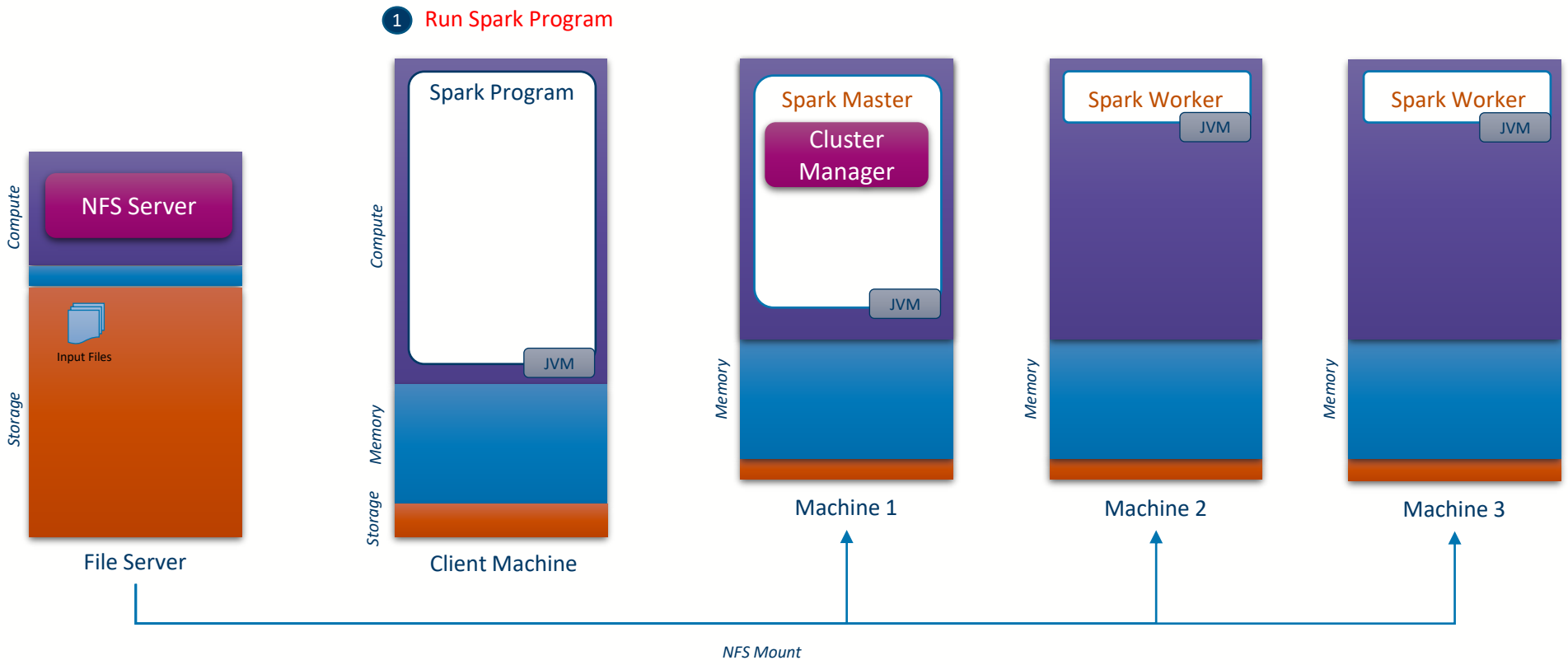
University of Glasgow

Spark Execution Phases

1. Client instantiates a SparkContext object (SC)
   - Responsible for I/O with the user, cluster manager service, and executors
   - Initialises the Driver, which registers the application with the Cluster Manager and gets a list of Executors
   - The Driver then is responsible for the Spark job
2. Client uses SC to create input RDDs
   - Partitioned/parallelised across cluster nodes
3. Client uses Transformations on the input/intermediate RDDs
   - Sequence of transformations creates a DAG of ops
   - Lazy evaluation ➔ No computation takes place so far on the cluster
4. Client uses an Action to print/store results
5. Action triggers submission of ops DAG to the DAGScheduler within the Driver
6. The DAGScheduler builds stages of tasks
   - Defines what will be executed where
   - Implemented as an event queue
   - Groups together tasks based on their dependencies
   - Submits the execution plan to the TaskScheduler within the Driver
   - Resubmits failed stages, if their output was lost
7. The TaskScheduler:
   - Ships tasks to Executors (serialised RDD lineage + transformations)
   - Monitors the execution/progress of their computation
     - Launches tasks on Executors, according to resource and locality constraints
     - Decides where to run each task
     - Deals with stragglers (speculative execution)
   - Returns the result to the DAGScheduler (as a series of events)

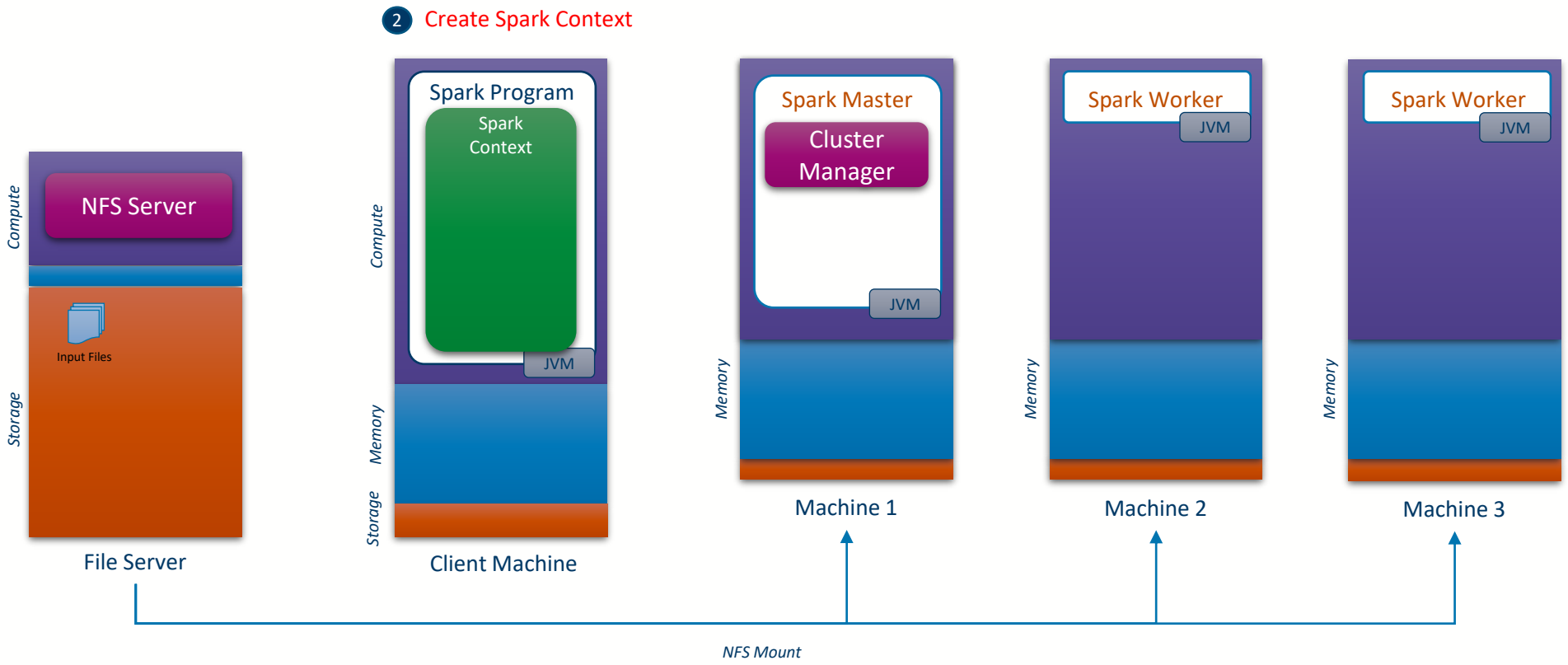
University of Glasgow

Job execution anatomy



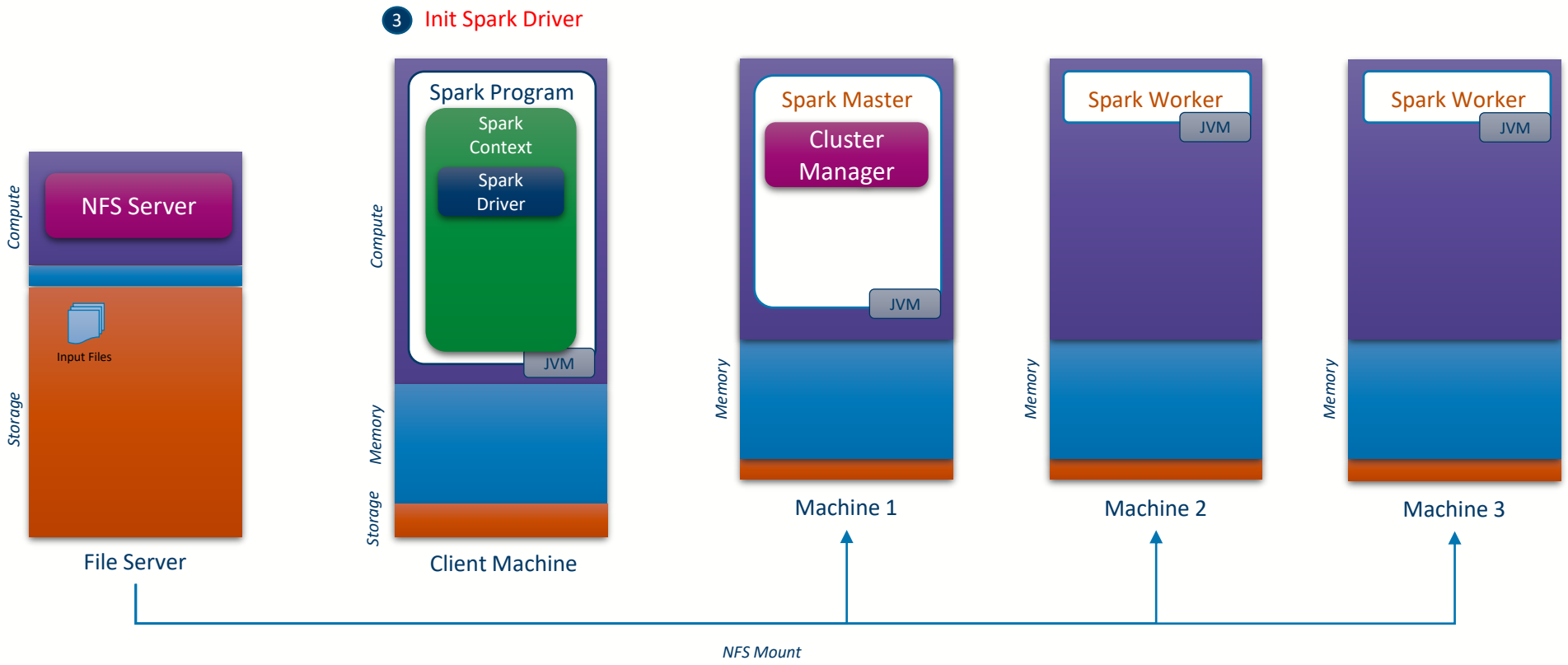
University of Glasgow

Job execution anatomy



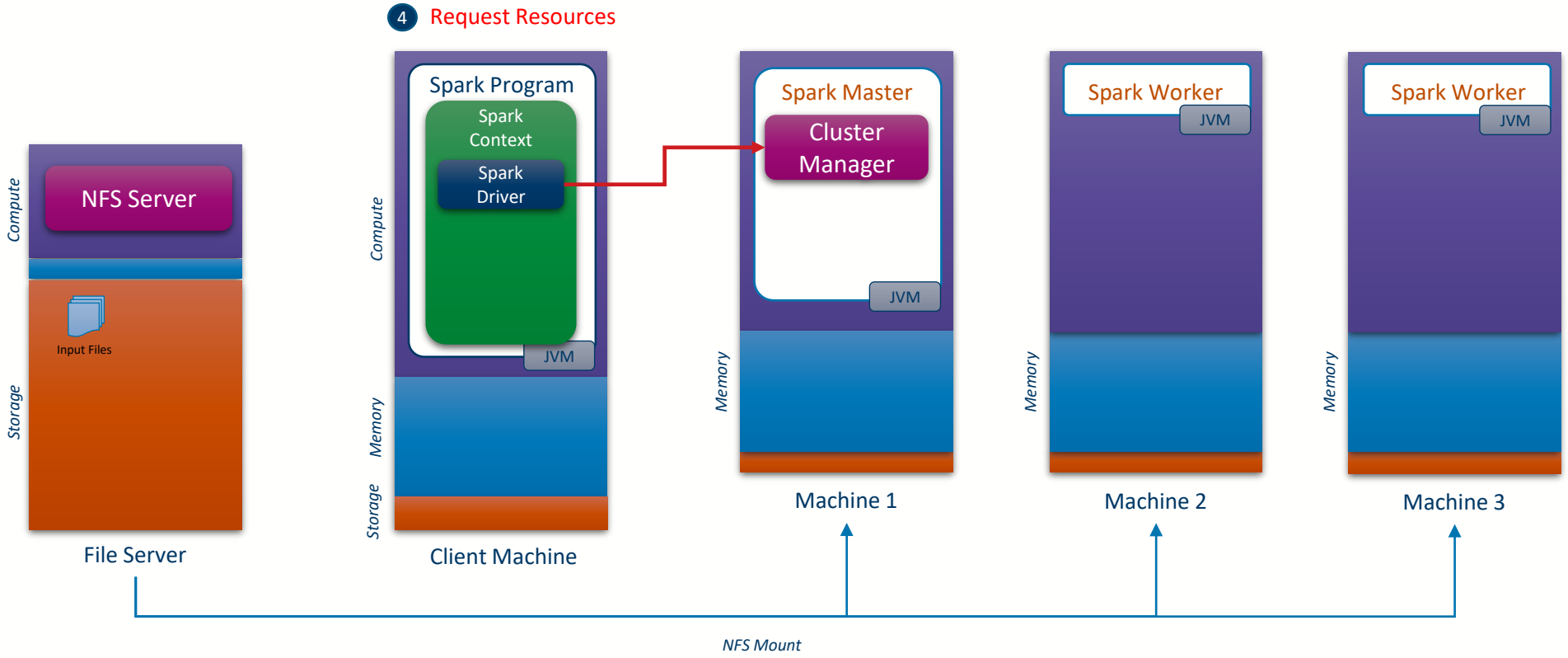
University of Glasgow

Job execution anatomy



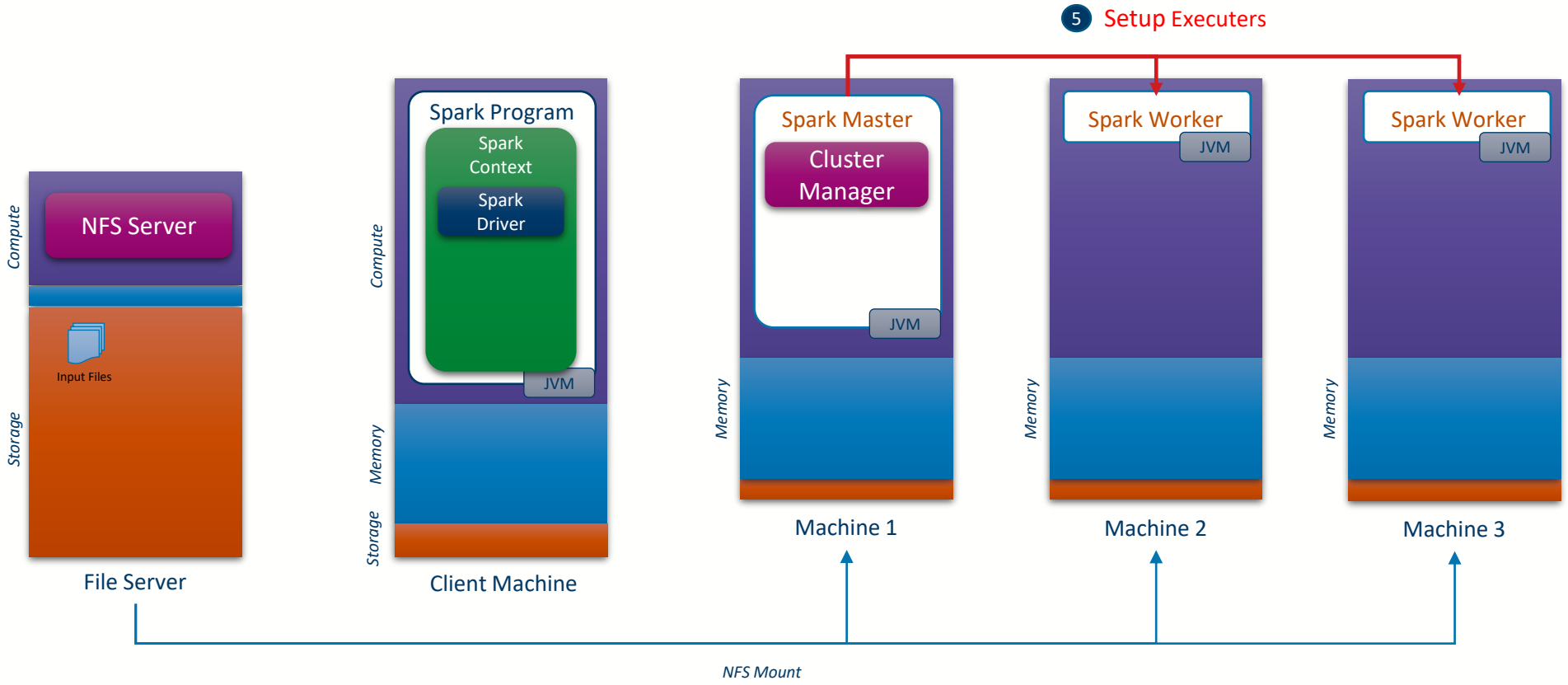
University of Glasgow

Job execution anatomy



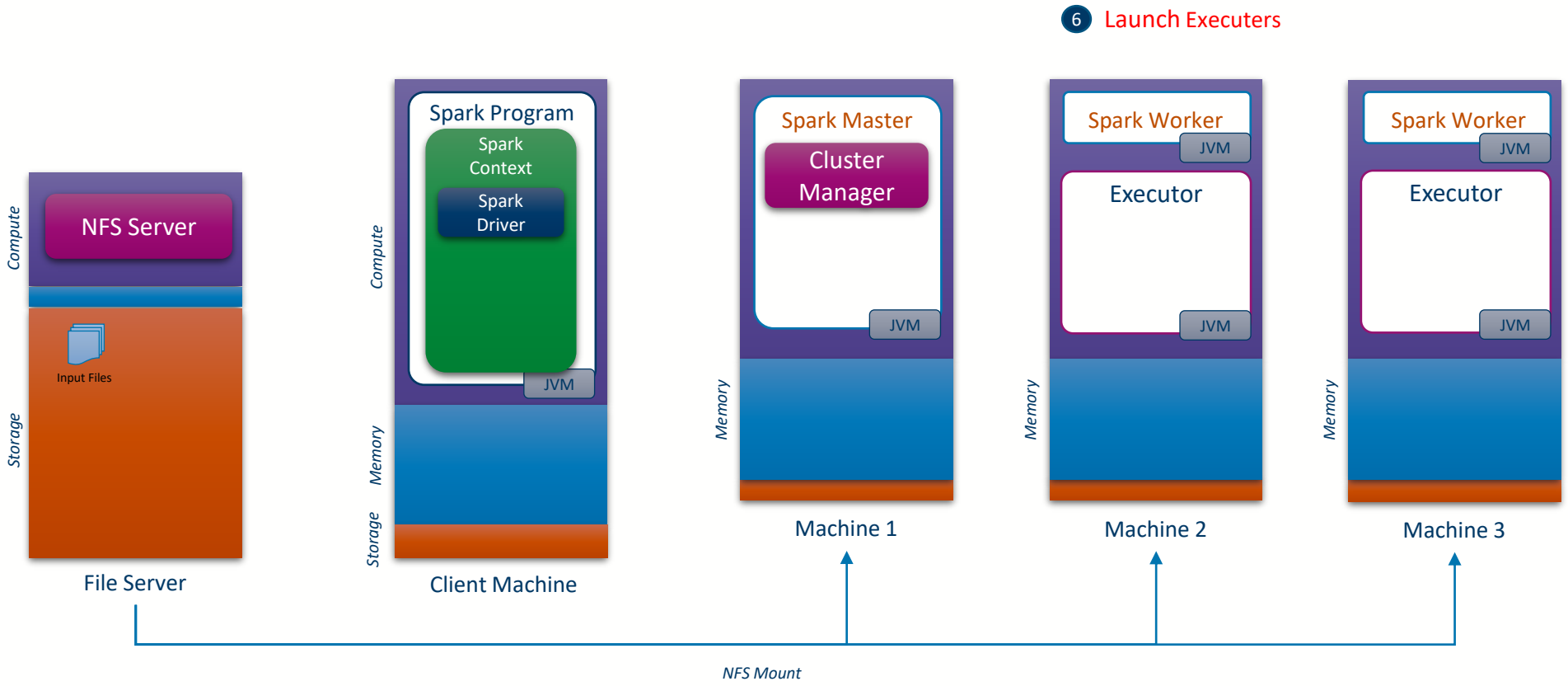
University of Glasgow

Job execution anatomy



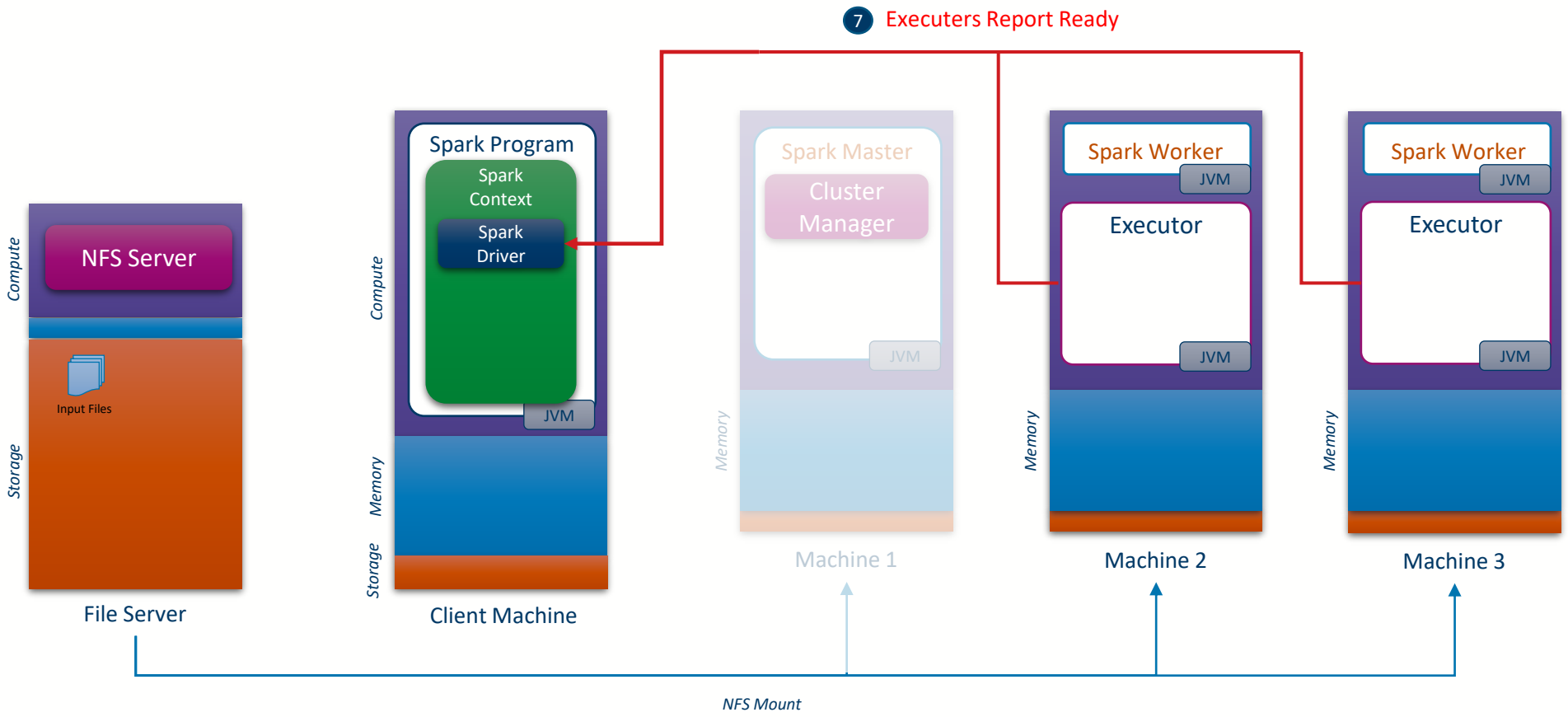
University of Glasgow

Job execution anatomy



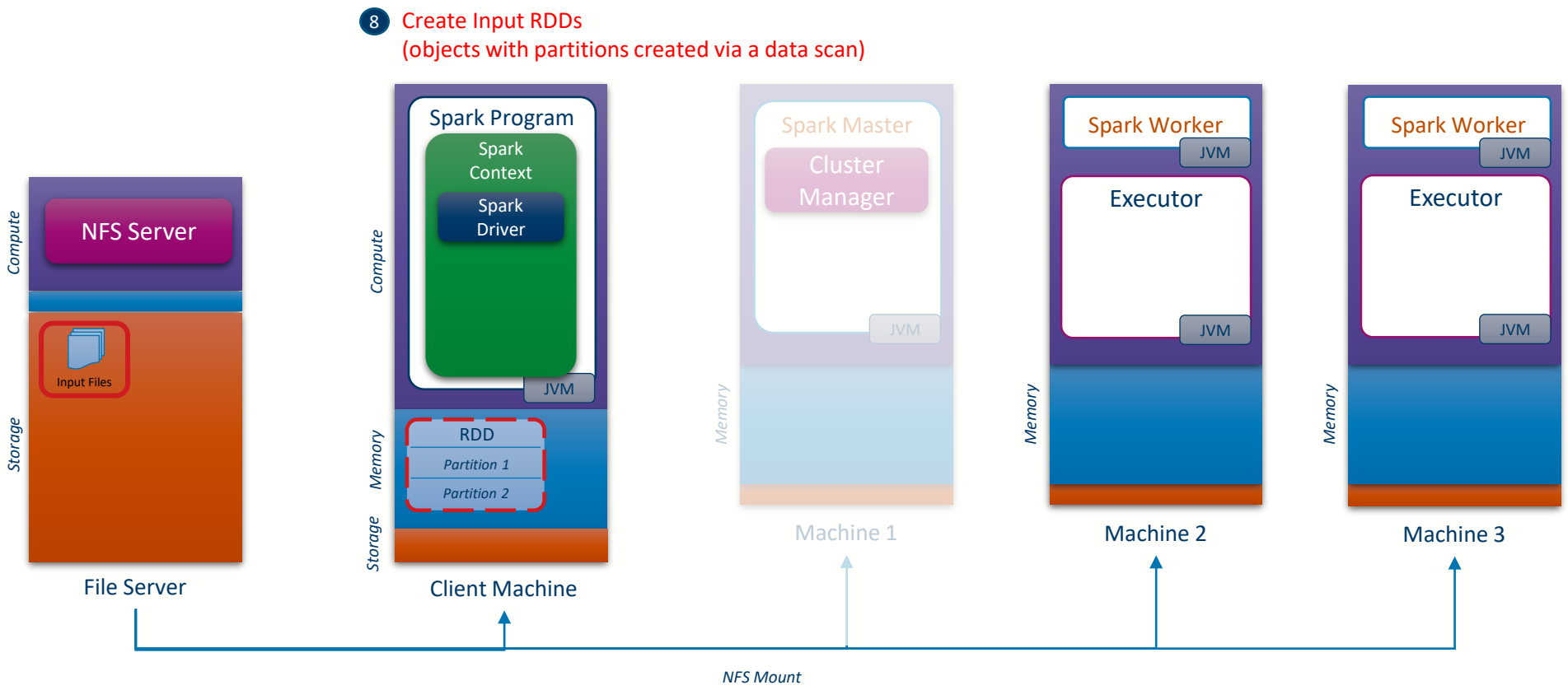
University of Glasgow

Job execution anatomy



University of Glasgow

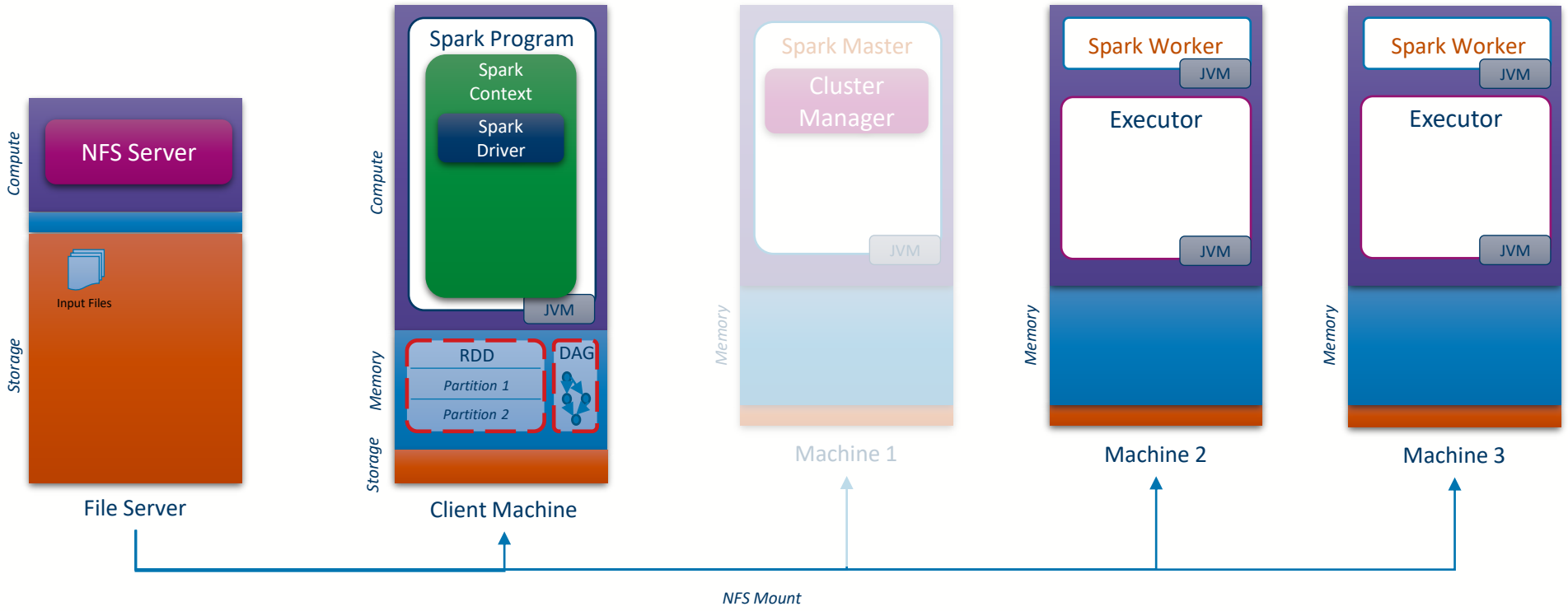
Job execution anatomy



University of Glasgow

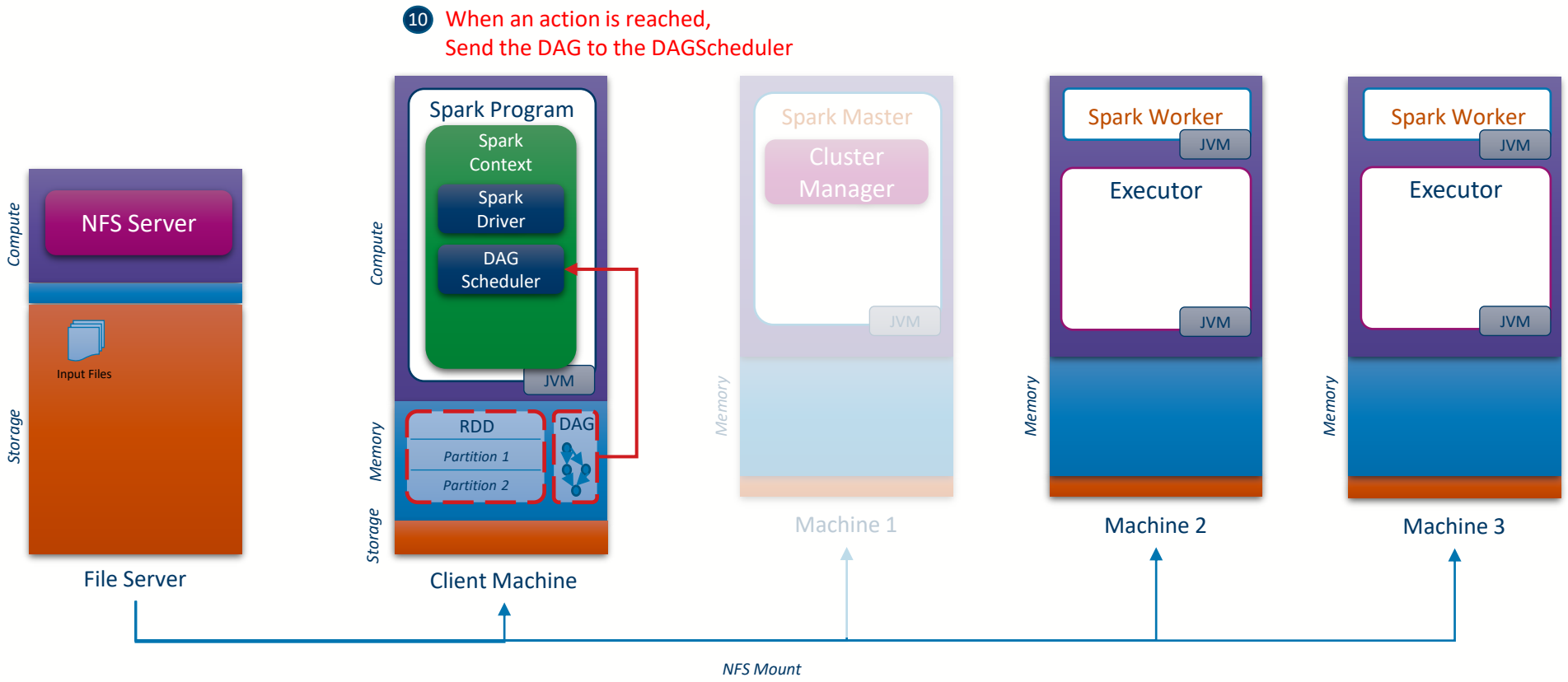
Job execution anatomy

9 Execute Transformations in lazy mode, Incrementally building the DAG



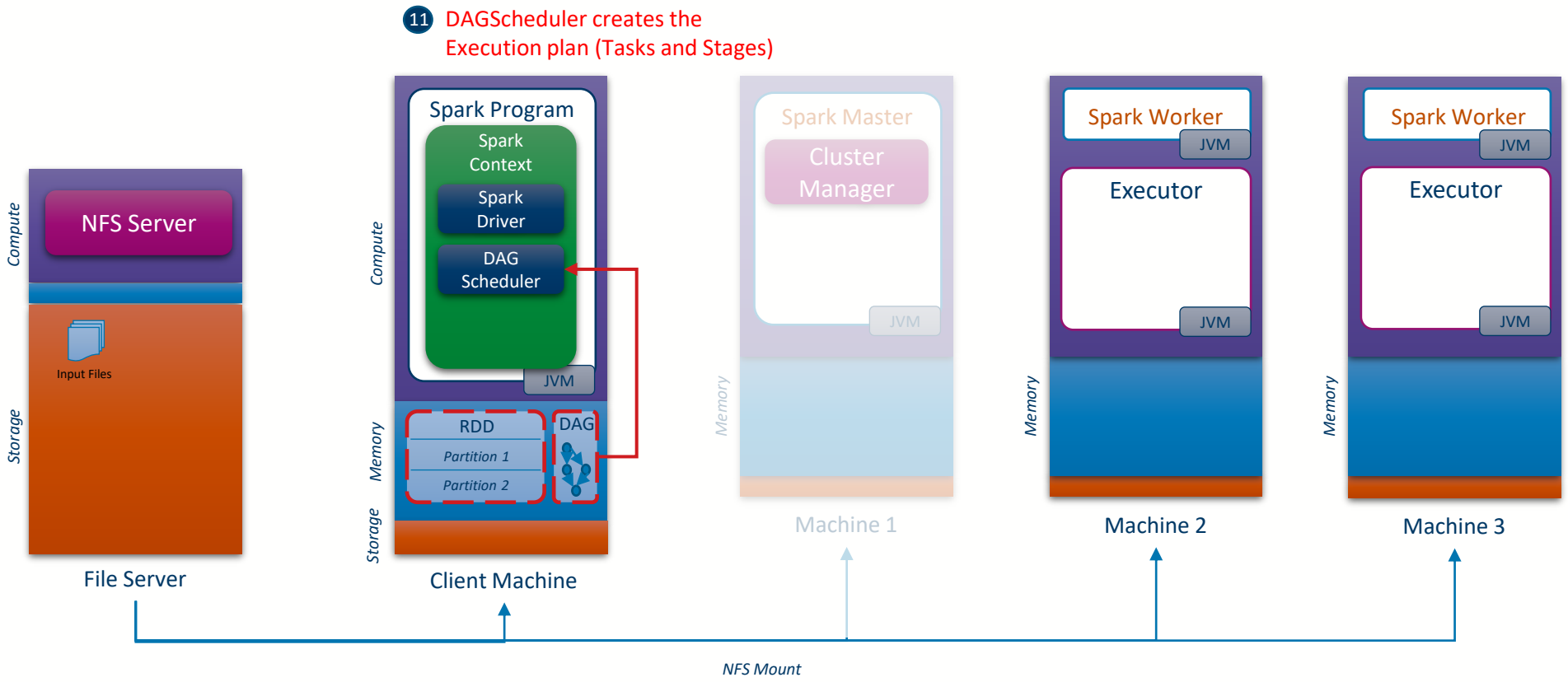
University of Glasgow

Job execution anatomy

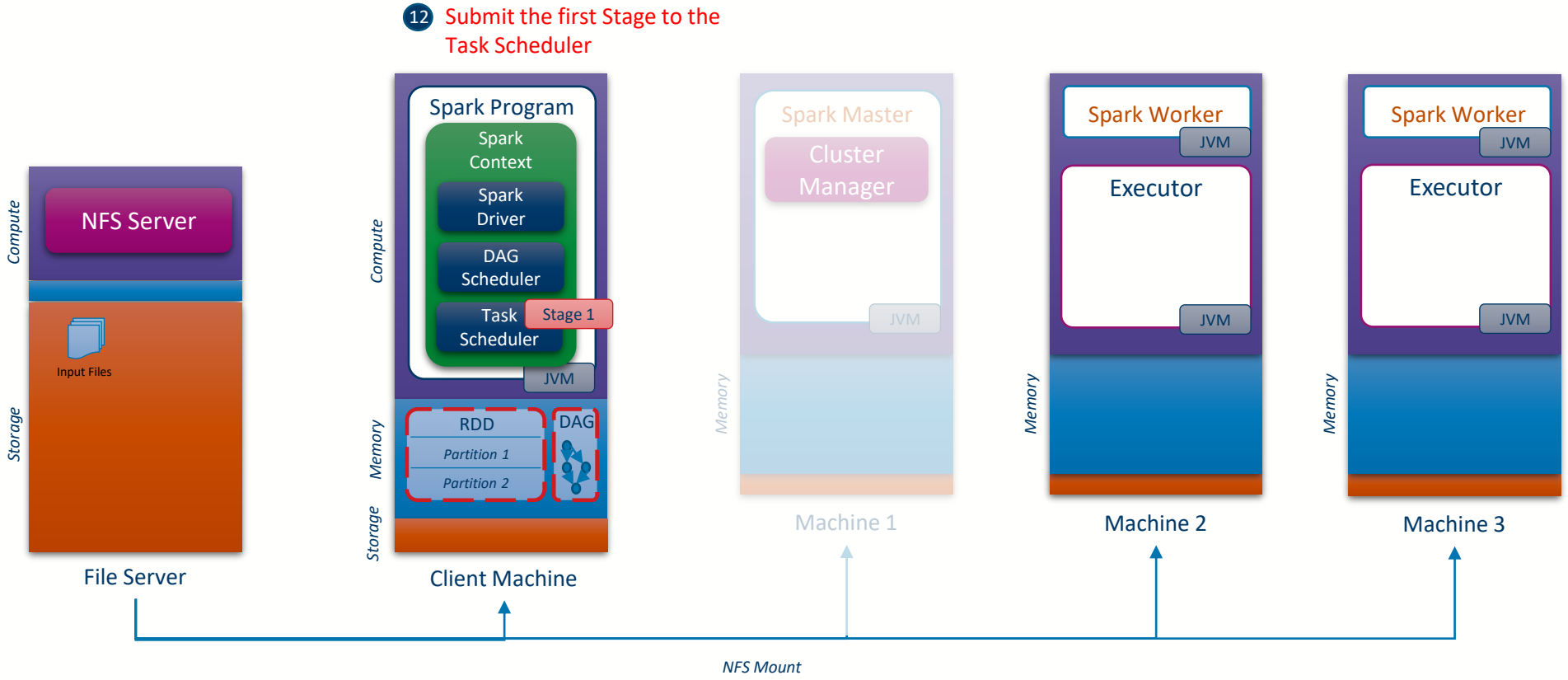


University of Glasgow

Job execution anatomy

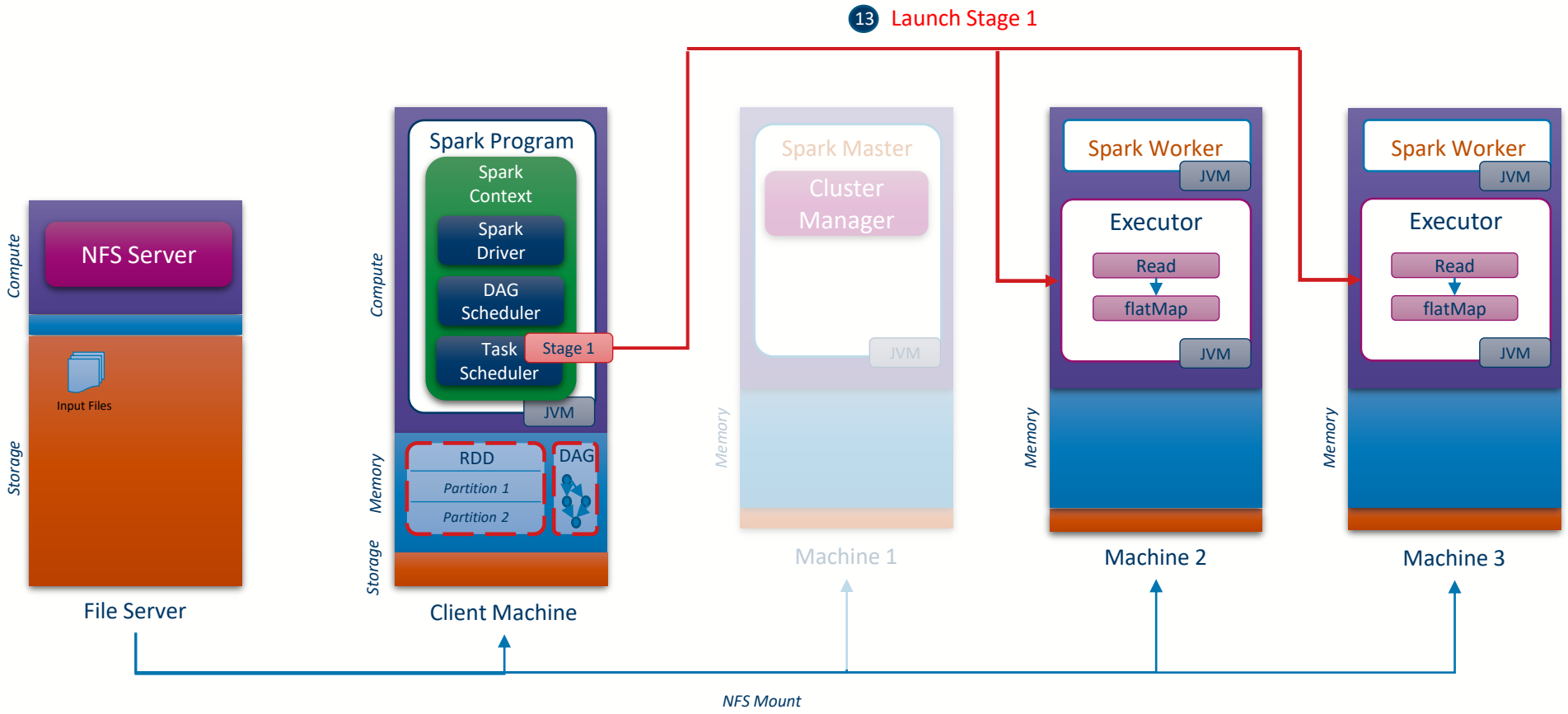


Job execution anatomy



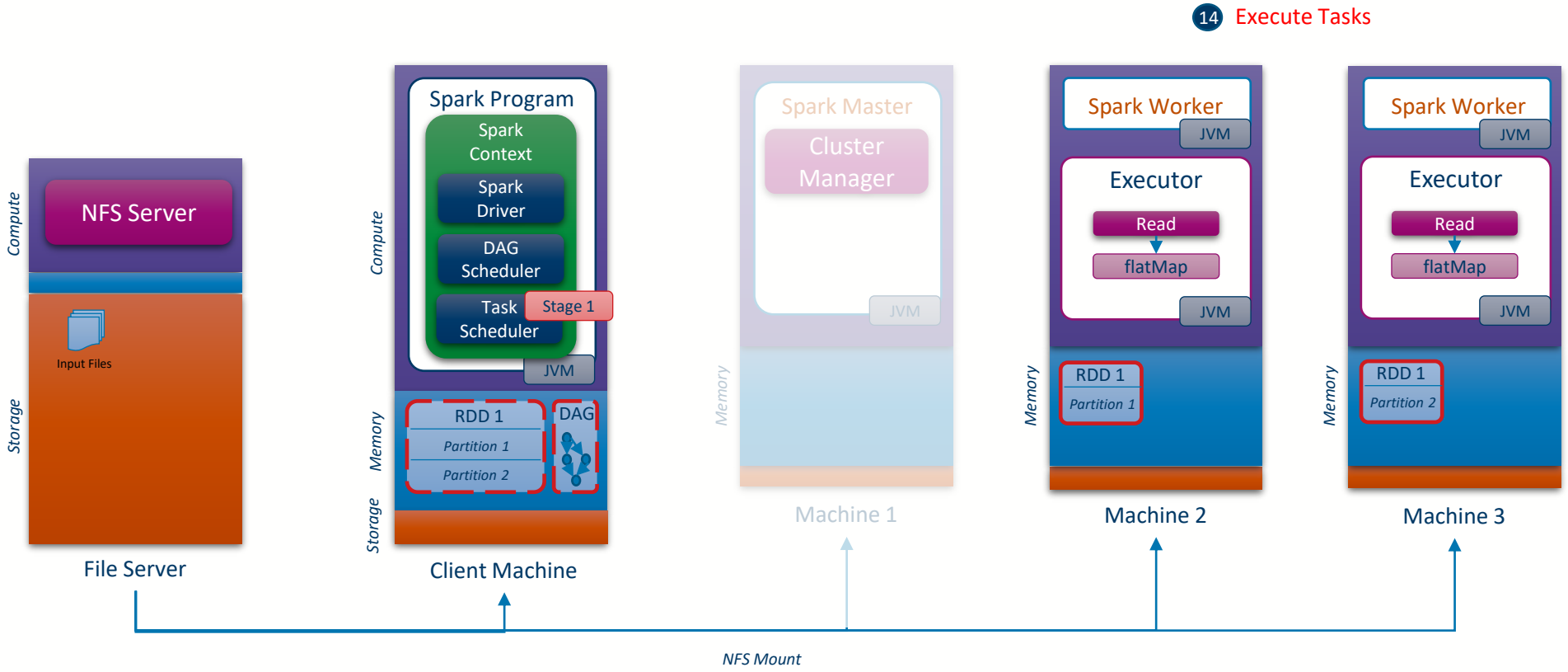
University of Glasgow

Job execution anatomy



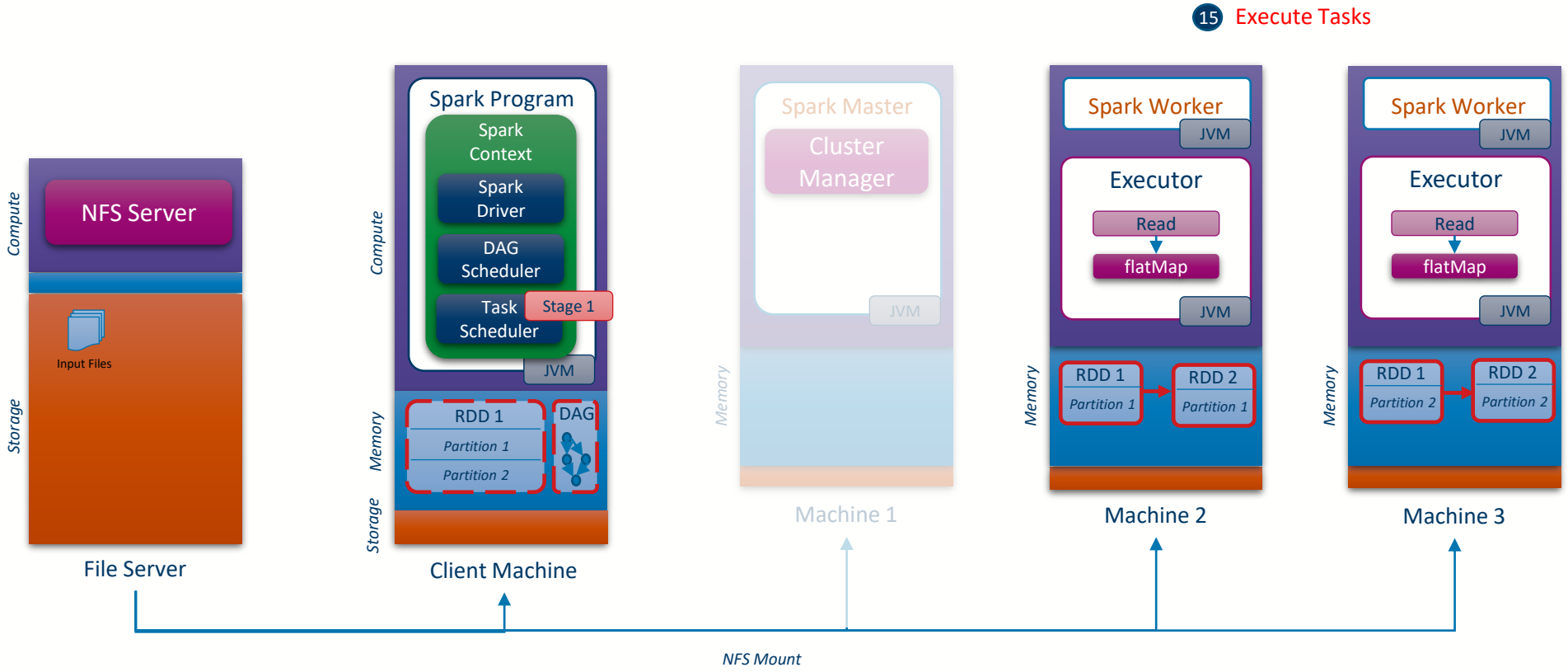
University of Glasgow

Job execution anatomy



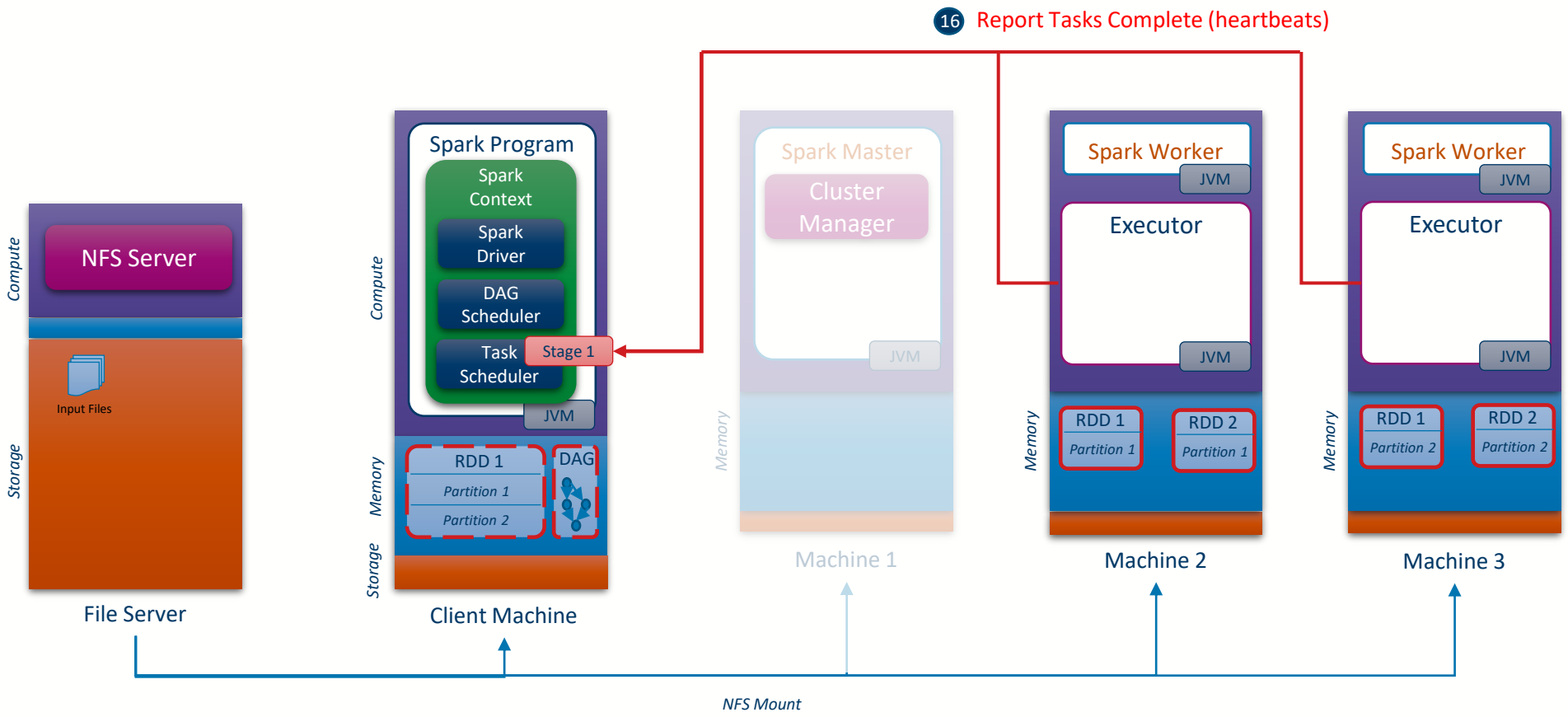
University of Glasgow

Job execution anatomy



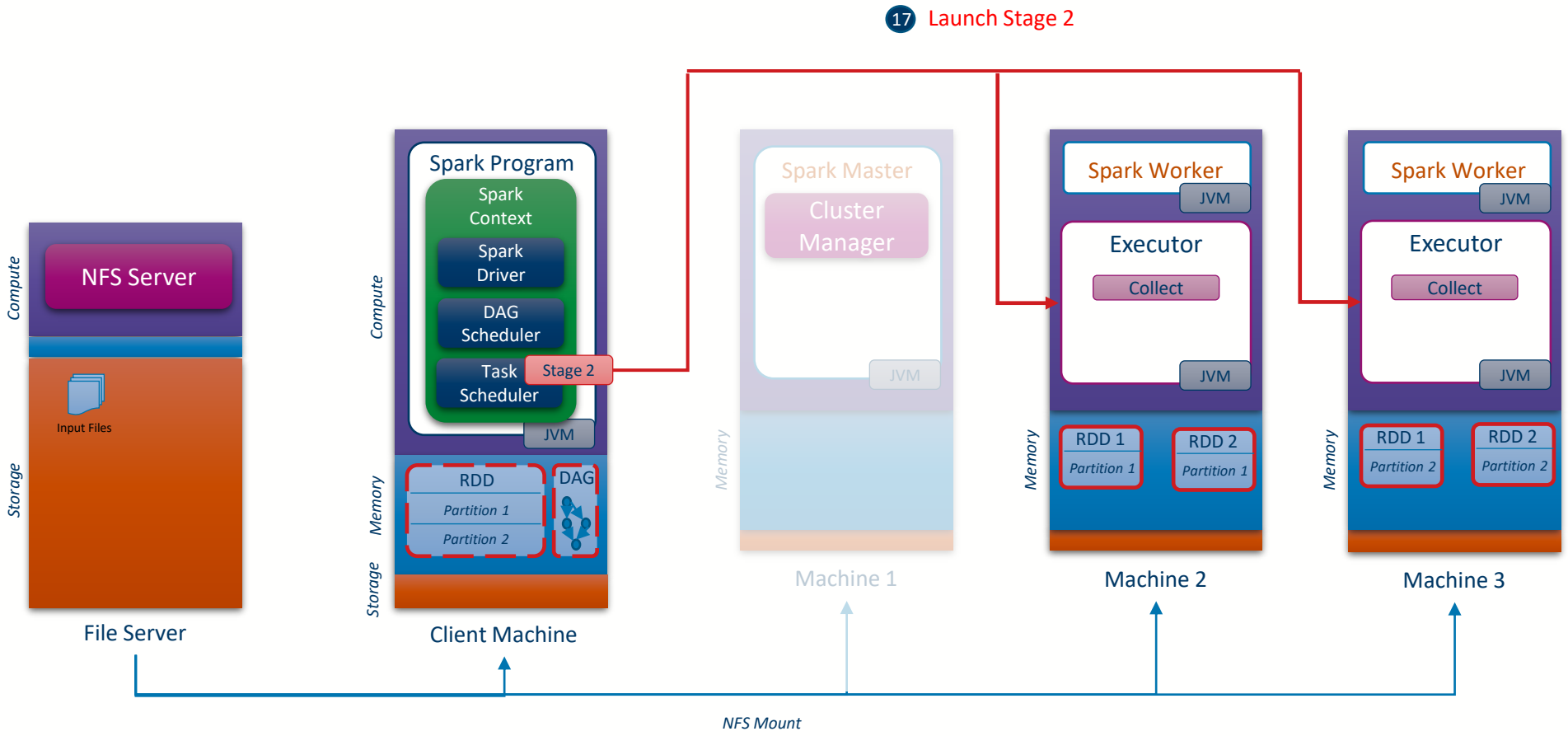
University of Glasgow

Job execution anatomy



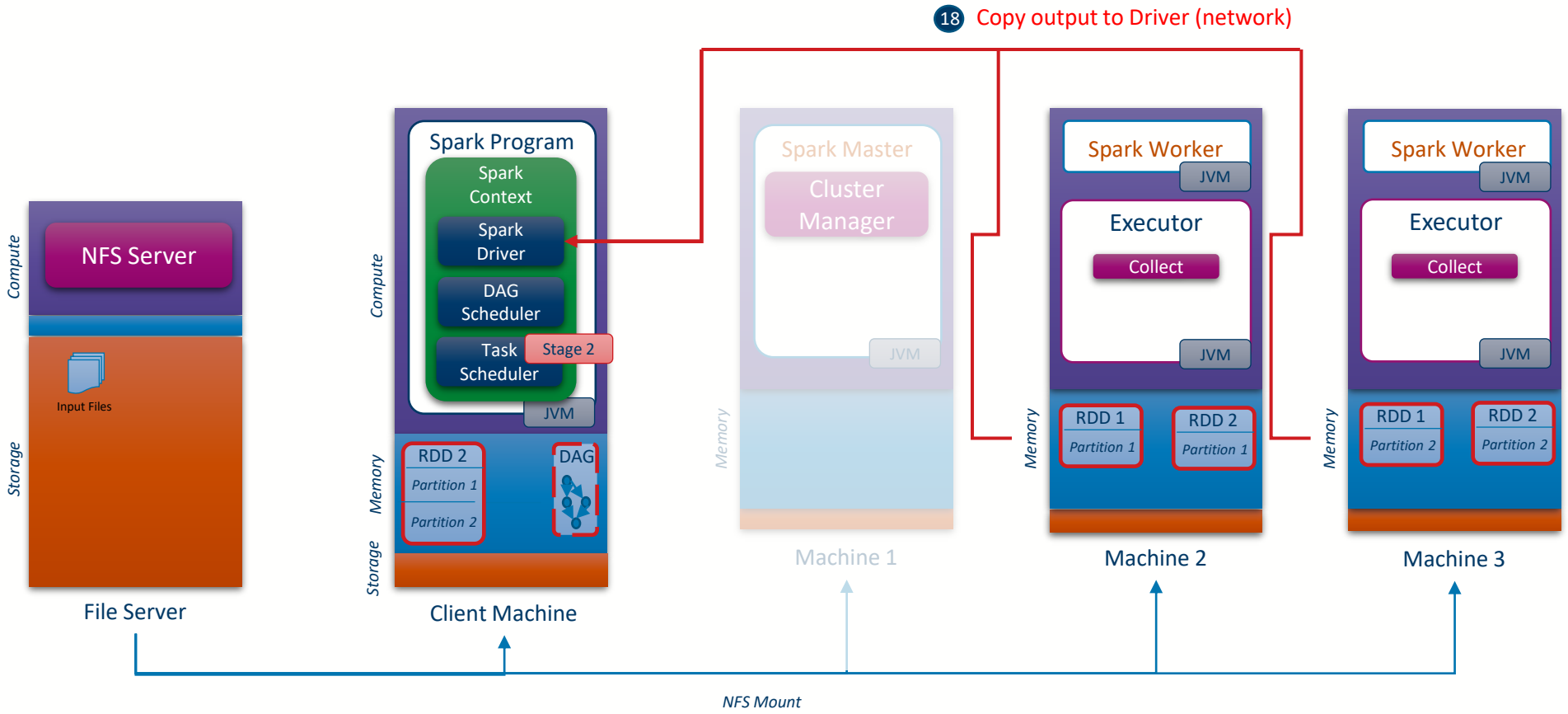
University of Glasgow

Job execution anatomy



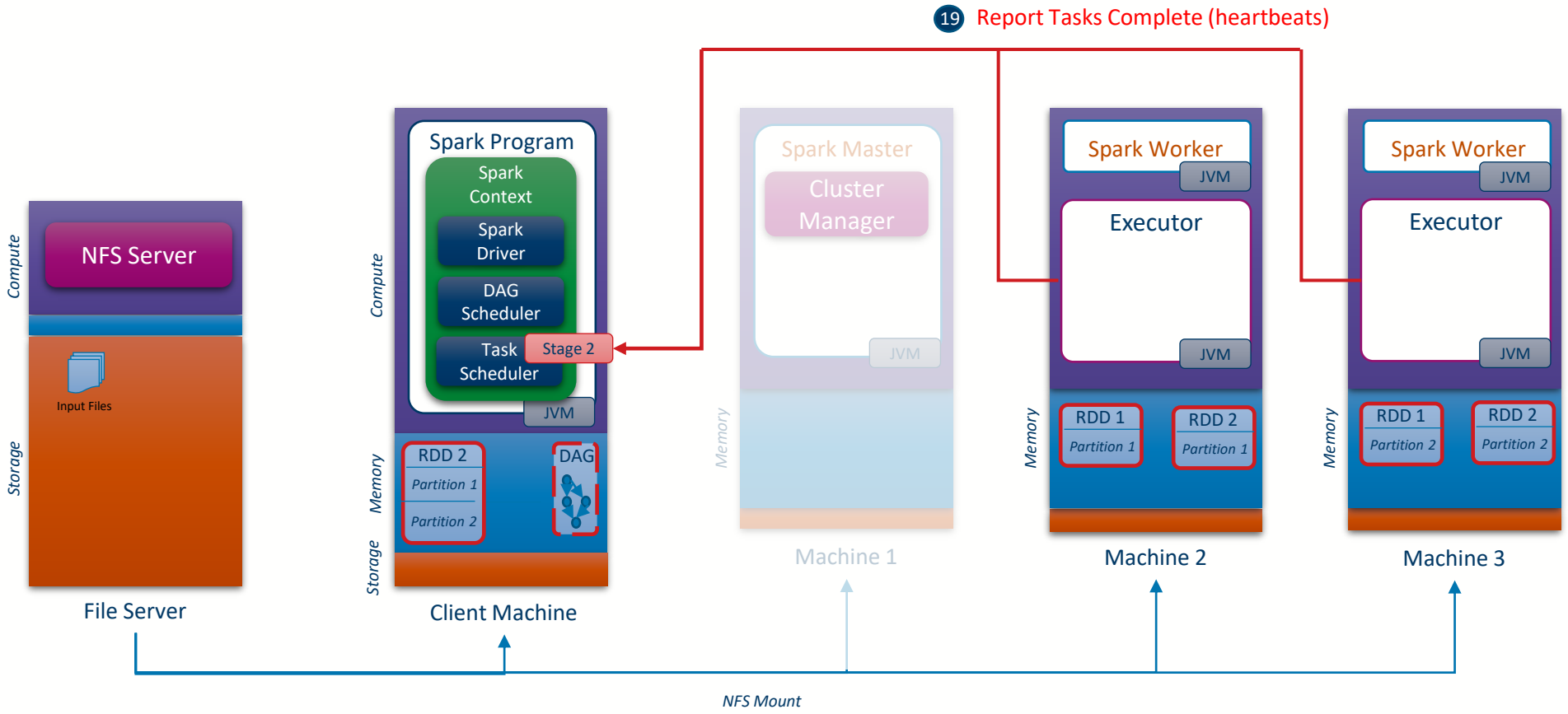
University of Glasgow

Job execution anatomy

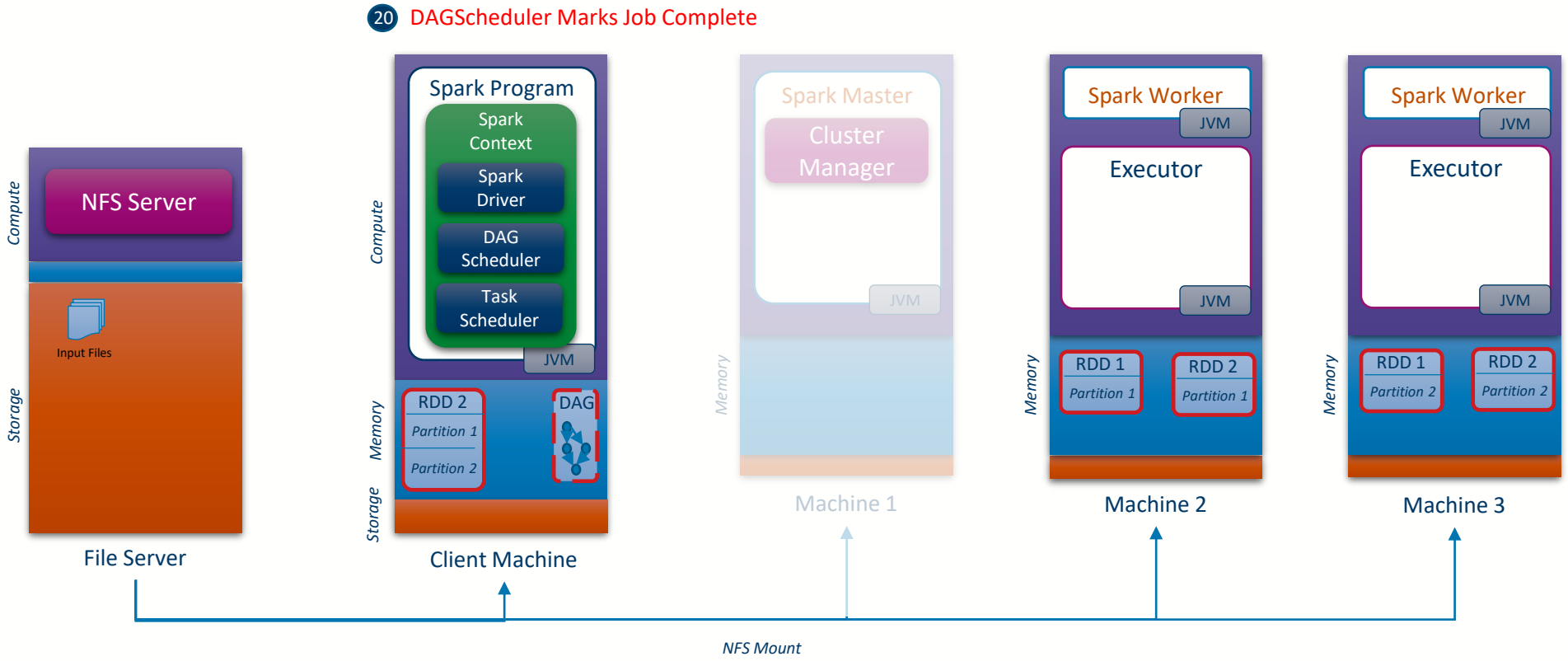


University of Glasgow

Job execution anatomy

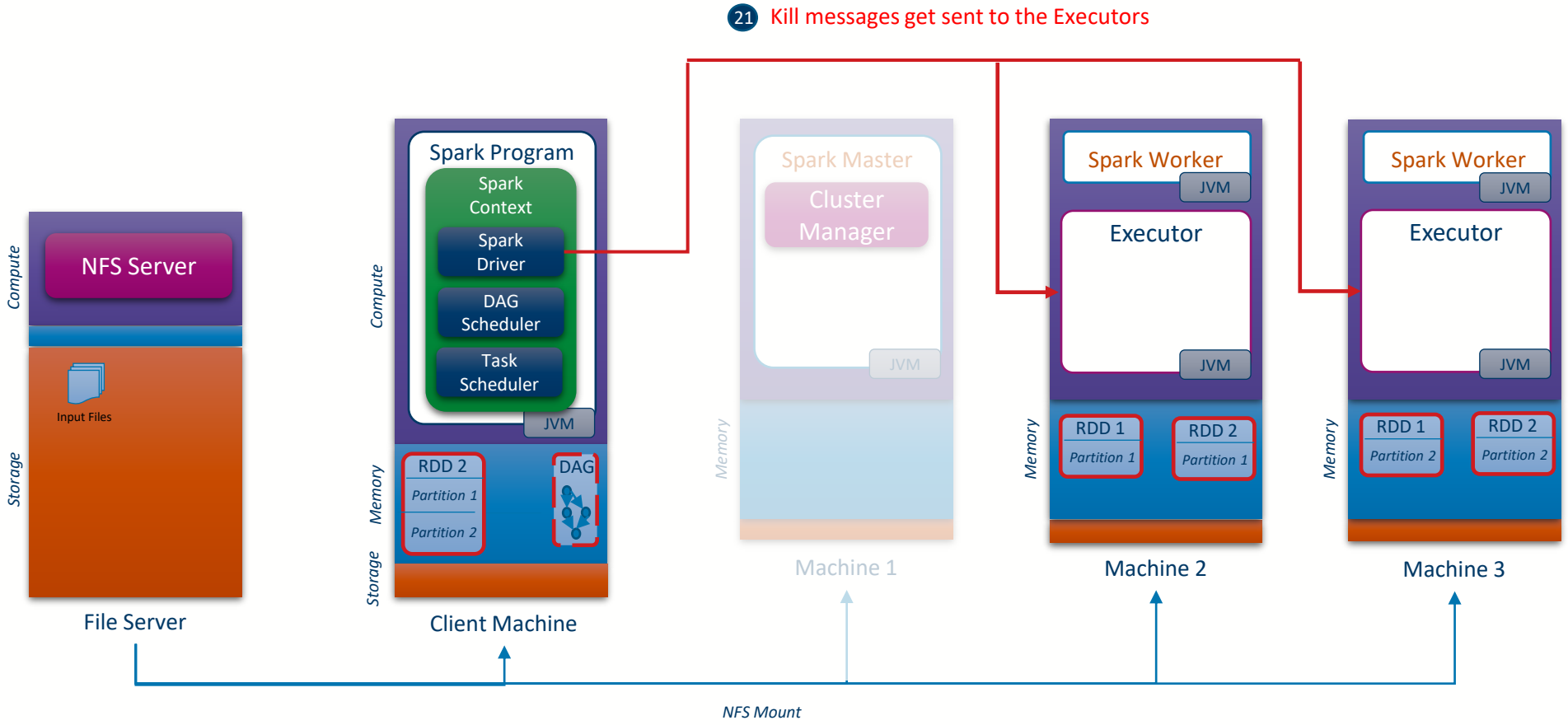


Job execution anatomy



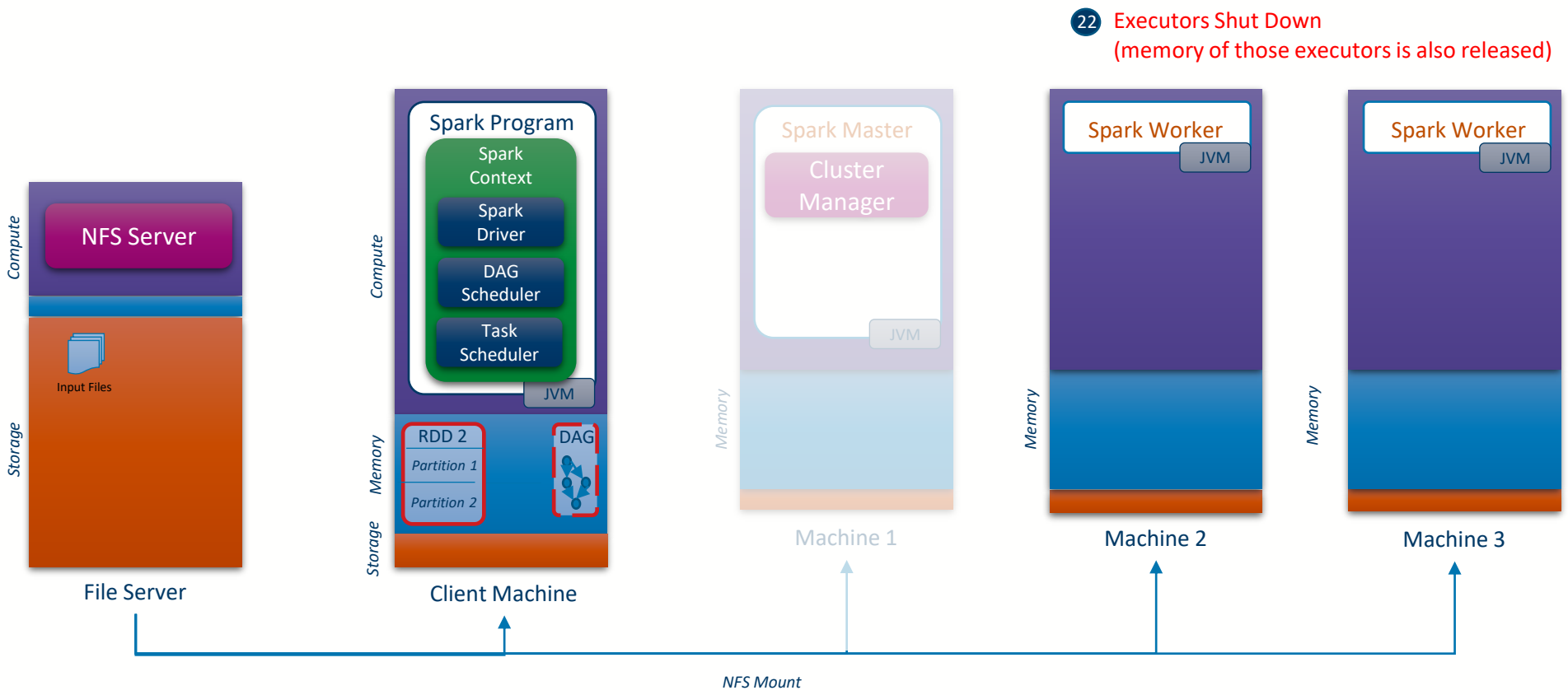
University of Glasgow

Job execution anatomy



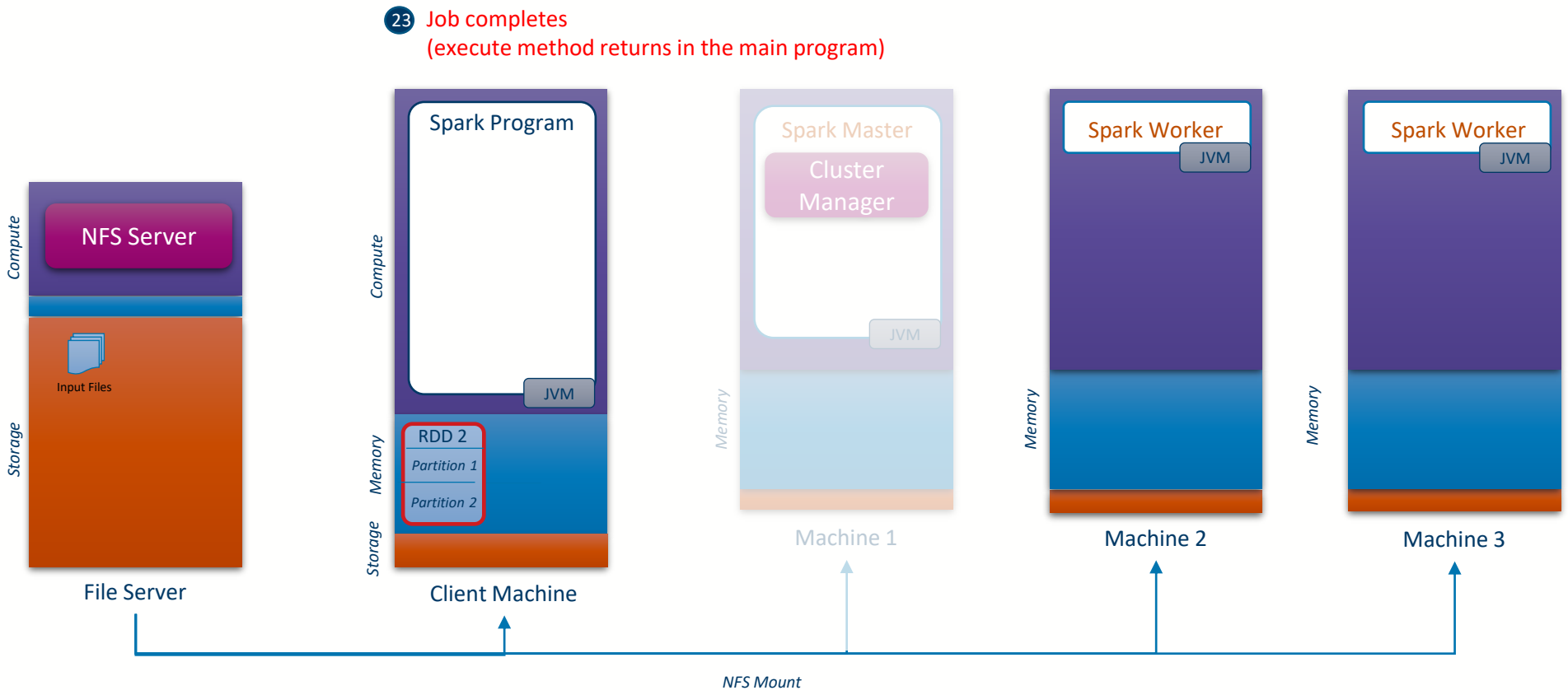
University of Glasgow

Job execution anatomy



University of Glasgow

Job execution anatomy



Course Coordinator
Dr. Richard McCreadie

Email: richard.mccreadie@glasgow.ac.uk
Room: SAWB 304

#UofGWorldChangers
@UofGlasgow